

ACTIVE LQR MULTI-AXLE-STEERING METHOD FOR IMPROVING MANEUVERABILITY AND STABILITY OF MULTI-TRAILER ARTICULATED HEAVY VEHICLES

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ABSTRACT—Directional performance and highway stability are two important aspects that need to be considered in development and design of a heavy articulated vehicles. To improve the maneuverability and stability of a multi-trailer articulated heavy vehicle (MTAHV), an active linear quadratic regulator (LQR) multi-axle-steering method is designed and examined. First, a linear yaw-plane model with four-degree-of freedom (4-DOF) for MTAHV is built and validated. Thus, a reference model supplying the desired state responses is introduced. Then, an active control algorithm of multi-axle-steering for the rear axles of tractor and full-trailer is investigated, and a LQR controller is proposed based on the linear vehicle model to make the control variables track the desired state responses. The control strategy concentrates on keeping the actual yaw rate and side-slip angle follow the steady-state yaw rate and zero side-slip angle. Finally, the effectiveness of the designed approach on enhancing the maneuverability and stability of the MTAHV have been validated through the simulations of the low-speed 360° roundabout and the single lane-change maneuver with high speed, respectively. The method has a certain reference value for improving the active safety of the MTAHV.

KEY WORDS : Directional performance, Highway stability, Maneuverability, Multi-trailer articulated heavy vehicle, Linear quadratic regulator, Active multi-axle-steering, Path following off-tracking, Rearward amplification

1. INTRODUCTION

In comparison to using an articulated heavy vehicle (AHV) with a single-trailer, the utilization of a MTAHV can economize 29 % transportation cost, diminish 44 % driven distance, reduce 32 % fuel use and greenhouse gas levels, and decrease 40 % tire wear and road damage (Ni and He, 2018; Woodroffe and Ash, 2001). A powered unit and two or multiple towed units form a MTAHV. The front-most unit, namely, a tractor, and the towed units is called trailers. Separate units are coupled together with a mechanical connections between each other at the hinge position, consisting of fifth wheels, pintle hitches, and dollies (Islam *et al.*, 2015). In spite of the benefits, due to their complicated structure, larger sizes, and higher center of gravity (CG) position, MTAHVs usually exhibit unacceptable directional performance in the respects of low high-speed lateral stability characteristics and poor low-speed mobility (Regehr *et al.*, 2009; Wideberg and Dahlberg, 2009; Ding *et al.*, 2013; Wang *et al.*, 2015). Firstly, MTAHVs frequently display three dangerous motion patterns at high speeds, consisting of

roll-over, trailer swing, and jack-knifing (Brown *et al.*, 2020; Palkovics and Fries, 2001; Fancher and Winkler, 2007). These three dangerous motion patterns may cause deadly traffic accidents. Secondly, when driving on urban streets and local roads at low speeds, MTAHVs have poor path-tracking performance (He *et al.*, 2010; Islam and He, 2009), which may affect the rate of damage to the road and tires under tight cornering to a certain extent (Odhams *et al.*, 2008). Furthermore, poor path-tracking of MTAHVs increases safety problems arising from the traffic environment and damage of road infrastructure (Rangavajhula and Tsao, 2007). Given these challenges, the development and adoption of effective advanced safety technologies for these large vehicles will have multiple effects on environmental, traffic safety benefits, and economic, allowing commercial vehicle transportation and road systems to profit. At the same time, it can mitigate environmental impacts, lower transportation costs, reduce road-surface damage, alleviate the unfavorable danger of increased traffic jams and traffic crashes (Ni and He, 2018).

The lateral dynamic of MTAHVs can usually be measured in respects of lateral stability and maneuverability (He *et al.*, 2017). To characterize the lateral stability evaluation

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efficiently under high-speed condition, a unified criterion is recognized as the rearward amplification (RWA), which is determined as the ratio of the maximum lateral acceleration of the rearmost trailer's CG to that of the powered unit, in a single lane-change evasive test at high speeds (Fancher and Winkler, 2007; Islam *et al.*, 2012; ISO 14791, 2000; Wang and He, 2016). The motions of the trailer swing and roll-over are connected directly to the lateral acceleration amplification experienced at CG of the semi-trailer. On the other hand, to characterize the low-speed maneuverability evaluation effectively, a unified criterion is presented as path following off-tracking (PFOT), which is recognized as the peak radial path offset, between the centre of tractor's front-axle and that of the rearmost semi-trailer's rear-axle, in a PFOT test manoeuvre specified (Islam *et al.*, 2012). A 90° intersection turn (Prem *et al.*, 2001) and a 360° roundabout path-tracking manoeuvre at low speeds (Tsao *et al.*, 2006) can be used individually or together for the PFOT test manoeuvre.

To reduce highway accidents, improve the handling stability at high speed and the accuracy of trajectory tracking for AHVs, a variety of effective active steering control methods have been investigated by researchers. The active steering control methods can be categorized into three groups.

In the first group, some useful active trailer steering (ATS) strategies acting on the single trailer have been proposed on the semi-trailer vehicles (He *et al.*, 2010; Odhams *et al.*, 2008; Islam *et al.*, 2012; Prem and Atley, 2002; Cheng and Cebon, 2008), many researches have been contributed to developing ATS systems for MTAHVs (Rangavajhula and Tsao, 2007; Rangavajhula and Tsao, 2008; Ding and He, 2011; Keldani and He, 2019), and the ATS system can also enable MTAHVs to coordinate curved trajectories and improve path-tracking performance at low speeds (He *et al.*, 2017). Furthermore, the ATS system can enhance the anti-roll stability of MTAHVs, which is verified by the simulation results (Cheng and Cebon, 2008).

Rangavajhula *et al.* (2007) proposed a LQR-based ATS controller, the minimum Rearward Amplification (RWA) was applied as a control criterion to achieve the control purpose of eliminating off-tracking. The same approach is also used to indirectly minimize off-tracking by minimizing the RWA at low speeds (Rangavajhula and Tsao, 2008). To enhance the roll stability of a tractor semi-trailer, an active semi-trailer steering control model based on the LQR controller was proposed, with minimizing the combination of track deviation and the trailer's lateral acceleration (Cheng and Cebon, 2008). An ATS systems based on LQR theory for AHVs was presented, which was designed with the criteria of minimizing the RWA ratio of the truck-trailer system and minimizing the PFOT value, and the LQR strategy was demonstrated to improve the lateral stability and the path tracking (Islam *et al.*, 2012). A LQR-based ATS controller was developed to improve the lateral stability of a B-train

double, which was designed by minimizing the performance index, the simulation results suggest that the LQR-based controller can restrain the lateral acceleration and simultaneously maintain an RWA ratio close to 1 (Keldani and He, 2019).

In the second group, the improvement of the tractor is key since some researchers believe that the driver's steering control relies strongly on his/her perception and feeling of the behavior of tractor unit. The second group was concerned mainly with active tractor steering control to enhance the tractor handling behavior. Palkovics and El-Gindy (1996) introduced an LQR to minimize the yaw rate and side-slip angle of tractor, etc. by active steering. El-Gindy *et al.* (2001) followed the idea of minimization using the LQR controller.

In the third group, it is more reasonable and practical economically to design an active dolly steering system than a trailer steering system, which improves the high-speed dynamic stability of the A-double owing to the small size of the dolly unit (Kati *et al.*, 2021). Kati (2018) has proposed an active dolly steering system where both axles of the dolly were considered steerable. To enhance the high-speed lateral stability of an A-double combination vehicle, it is proposed a feedback-feed-forward control strategy using active dolly steering (Kati *et al.*, 2021). Rahimi *et al.* (2021) developed an active dolly and trailer steering control strategy, which results in a more comprehensive and reliable autonomous driving control solution.

As demonstrated in the above studies, much effort and progress has been accomplished in active control steering for tractor, trailer or dolly steering. Currently, research on active steering of AHVs has mainly focused on improving the handling behavior of the tractor or trailer. Few studies have proposed to improve the maneuverability and stability of the whole tractor-trailer combination through simultaneous active steering control of the tractor and trailer, and little research has presented an active multi-axle-steering control strategy acting on the rear axles of the tractor and full-trailers for MTAHVs. It can be inferred that the stability of the long combination vehicles (LCVs) is satisfactory if the handling indexes of both the tractor and trailers are controlled within acceptable limits.

The LQR-based controller has been applied to AHVs active steering control in many cases. However, it appears that the minimization method has an obvious advantage over low-speed maneuverability in improving high-speed stability, the algorithm of minimizing yaw rate may not be the best choice, and the tracking the desired yaw rate will have a better effect (Tabatabaei Oreh *et al.*, 2013; Kim *et al.*, 2016). In this paper, instead of using the approach of minimization, there is a critical need for devising a control method which can make the actual behaviour of MTAHVs track a desired behaviour. The stability control strategy concentrated mainly on making the actual yaw rate continuously keep tracking the desired value and limiting the side-slip angle of the CG as much as possible. So that, based on the strategy of tracking

the steady-state yaw rate and zero side-slip angle, an active LQR multi-axle-steering tractor and full-trailer method is developed.

In this study the contributions are as follows: (1) To eliminate the off-tracking and enhance the stability of MTAHV, an active control algorithm of multi-axle-steering for the tractor's rear axles and full-trailer is investigated. (2) Instead of using the idea of minimization, a LQR controller is proposed based on the linear 4-DOF yaw plane model of MTAHV to make the control variables keep following the desired responses. The control strategy concentrates on keeping the actual yaw rate and side-slip angle follow the steady-state yaw rate and zero side-slip angle. (3) The effectiveness of the proposed approach on enhancing the maneuverability and stability of a MTAHV has been simulated and discussed using the simulations of the low-speed 360° roundabout trajectory tracking manoeuvre and the high-speed single lane-change test, respectively.

The rest of the paper is arranged as follows: a linear yaw-plane model with 4-DOF and a nonlinear TruckSim model for the MTAHV are established in Section 2. This section also validates the accuracy of the linear vehicle model. In Section 3, an active LQR multi-axle-steering method is designed based on the strategy of tracking the steady-state yaw rate and zero side-slip angle, and a reference model supplying the desired state responses is presented. Computer simulations and comparative analysis are carried out in Section 4 to investigate the effectiveness of the method proposed to enhance the lateral stability and maneuverability. Finally, conclusions are given in Section 5.

2. VEHICLE SYSTEM MODELLING

In this section, a linear yaw-plane model with 4-DOF and a nonlinear TruckSim model for the MTAHV have been established, the linear model will be used to design the control system and the reference model, and the nonlinear model will be applied to the assessment of the control system.

2.1. Linear Yaw-plane Model with 4-DOF for the MTAHV

In this paper, the lateral dynamics of the MTAHV with double semi-trailer and one tractor structure is to be studied. Each units of the vehicle are connected by the fifth wheel. To derive the active multi-axle-steering tractor and full-trailers method, Figure 1 shows the linear simplified yaw-plane model with 4-DOF for MTAHV. The vehicle system of MTAHV is simplified laterally and individual axle group is represented by one wheel. The MTAHV model consists of three rigid bodies: the tractor, the first trailer, and a second trailer. For the linear MTAHV model, the motions regarded are the lateral velocity and yaw rate of the tractor, the articulation angles between the adjacent units of the MTAHV. The wheels on the front and rear axle of the tractor, the wheels on each trailer axle are totally steerable, which configure the active multi-axle-steering tractor and full-trailer. At the corresponding CG of each vehicle unit, the tractor, first semi-trailer, and second semi-trailer are fitted to three different body-fixed coordinate systems, respectively. To generate the simplified model, the assumptions have been given below:

- (1) The forward speed of the MTAHV remains unchanged;
- (2) The longitudinal forces and the aerodynamic force are both neglected;
- (3) The pitch and rolling motions are not considered;
- (4) The front wheel steering angle is used directly as the system input;
- (5) The relationship between the lateral force of the tire and the side-slip angle of the tire is linear (Islam *et al.*, 2012; Wu and Hai, 2003).

The MTAHV parameters are listed in Table 1. As shown in Figure 1, applying Newton's second law, the control equations of the tractor and double trailer are derived theoretically, respectively. The motion equations of the tractor are

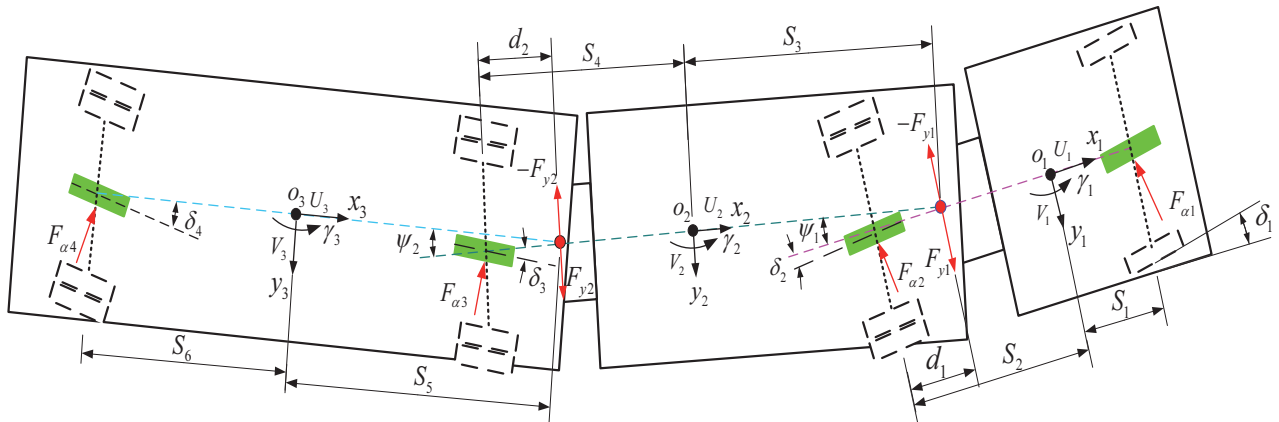


Figure 1. The linear simplified 4-DOF yaw-plane model for the MTAHV.

$$m_1 (\dot{V}_1 + U_1 \cdot \gamma_1) = F_{\alpha 1} \cos \delta_1 + F_{y1} + F_{\alpha 2} \cos \delta_2 \quad (1)$$

$$I_1 \dot{\gamma}_1 = F_{\alpha 1} \cos \delta_1 S_1 - F_{y1} (S_2 - d_1) - F_{\alpha 2} \cos \delta_2 S_2 \quad (2)$$

The motion equations of the first trailer are

$$m_2 (\dot{V}_2 + U_2 \cdot \gamma_2) = -F_{y1} \cos \psi_1 + F_{y2} + F_{\alpha 3} \cos \delta_3 \quad (3)$$

$$I_2 \dot{\gamma}_2 = -F_{y1} \cos \psi_1 S_3 - F_{y2} (S_4 - d_2) - F_{\alpha 3} \cos \delta_3 S_4 \quad (4)$$

Table 1. MTAHV parameters.

Symbol	Meaning	Symbol	Meaning
m_1	mass of the tractor, kg	β_1	side-slip angle of the tractor's CG, rad
m_2	mass of first trailer, kg	β_2	side-slip angle of the semi-trailer's CG, rad
m_3	mass of second trailer, kg	S_1	distance from the tractor's CG to the front axles, m
S_2	distance from the tractor's CG to the rear axles, m	d_1	distance from the first fifth wheel center to the tractor's rear axle, m
S_3	distance from the first trailer's CG to the first fifth wheel center, m	S_4	distance from CG to the rear axles of the first trailer, m
d_2	distance from the second fifth wheel center to the first trailer's rear axle, m	S_5	distance from the second trailer's CG to the second fifth wheel center, m
S_6	distance from CG to the rear axles of the second trailer, m	I_1	inertia moment of mass about the vertical axis of the tractor, kg·m ²
I_2	inertia moment of mass about the vertical axis of the first trailer, kg·m ²	I_3	inertia moment of mass about the vertical axis of the second trailer, kg·m ²
δ_1	steering angle of the tractor front wheels, rad	δ_2	steering angle of the tractor rear wheels, rad
δ_3	steering angle of the first trailer rear wheels, rad	δ_4	steering angle of the second trailer rear wheels, rad
C_1	total cornering stiffness of the front axle of the tractor, kN/rad	C_2	total cornering stiffness of the rear axle of the tractor, kN/rad
C_3	total cornering stiffness of the rear axle of the first trailer, kN/rad	C_4	total cornering stiffness of the rear axle of the second trailer, kN/rad
α_1	side-slip angle of front wheel of tractor, rad	α_2	side-slip angle of rear wheel of tractor, rad
α_3	side-slip angle of rear wheel of first trailer, rad	α_4	side-slip angle of rear wheel of second trailer, rad
$F_{\alpha 1}$	total lateral force acted on the front axle of the tractor, N	$F_{\alpha 2}$	total lateral force acted on the rear axle of the tractor, N
$F_{\alpha 3}$	total lateral force acted on the rear axle of the first trailer, N	$F_{\alpha 4}$	total lateral force acted on the rear axle of the second trailer, N
U_1	longitudinal velocity of the tractor, m/s	U_2	longitudinal velocity of the first trailer, m/s
U_3	longitudinal velocity of the second trailer, m/s	γ_1	yaw rate of the tractor, rad/s
γ_2	yaw rate of the first trailer, rad/s	γ_3	yaw rate of the second trailer, rad/s
ψ_1	articulation angle between the tractor and the first trailer, rad	ψ_2	articulation angle between the first trailer and the second trailer, rad
β_1	side-slip angle of the tractor body, rad	β_2	side-slip angle of the first trailer body, rad
β_3	side-slip angle of the second trailer, rad	F_{y1}	lateral reaction force at the first articulated point, N
F_{y2}	lateral reaction force at the second articulated point, N	V_2	lateral velocity of the first trailer, m/s
V_1	lateral velocity of the tractor, m/s	V_3	lateral velocity of the second trailer, m/s

The motion equations of the second trailer are

$$m_3 (\dot{V}_3 + U_3 \cdot \gamma_3) = -F_{y2} \cos \psi_2 + F_{\alpha 4} \cos \delta_4 \quad (5)$$

$$I_3 \dot{\gamma}_3 = -F_{y2} \cos \psi_2 S_5 - F_{\alpha 4} \cos \delta_4 S_6 \quad (6)$$

The lateral velocities equations between the two trailers can be expressed

$$\dot{V}_2 + S_3 \dot{\gamma}_2 = \dot{V}_1 - (S_2 - d_1) \dot{\gamma}_1 + U_1 (\gamma_1 - \gamma_2) \quad (7)$$

$$\dot{V}_3 + S_5 \dot{\gamma}_3 = \dot{V}_2 - (S_4 - d_2) \dot{\gamma}_2 + U_1 (\gamma_2 - \gamma_3) \quad (8)$$

Under the condition of small wheel angle, the lateral force of a tire and the side-slip angle of the tire is assumed to be a linear function, and the lateral tire force is expressed as

$$F_{\alpha i} = C_i \alpha_i, i = 1, 2, 3, 4 \quad (9)$$

The side-slip angle of each wheel is specified as a negative value, and the stiffness value is positive.

$$\alpha_1 = \beta_1 + \frac{S_1 \gamma_1}{U_1} - \delta_1 \quad (10)$$

$$F_{\alpha 1} = C_1 \left(\beta_1 + \frac{S_1 \gamma_1}{U_1} - \delta_1 \right) \quad (11)$$

$$\alpha_2 = \beta_1 - \frac{S_2 \gamma_1}{U_1} - \delta_2 \quad (12)$$

$$F_{\alpha 2} = C_2 \left(\beta_1 - \frac{S_2 \gamma_1}{U_1} - \delta_2 \right) \quad (13)$$

$$\alpha_3 = \beta_2 - \frac{S_4 \gamma_2}{U_2} - \delta_3 \quad (14)$$

$$F_{\alpha 3} = C_3 \left(\beta_2 - \frac{S_4 \gamma_2}{U_2} - \delta_3 \right) \quad (15)$$

$$\alpha_4 = \beta_3 - \frac{S_6 \gamma_3}{U_3} - \delta_4 \quad (16)$$

$$F_{\alpha 4} = C_4 \left(\beta_3 - \frac{S_6 \gamma_3}{U_3} - \delta_4 \right) \quad (17)$$

Through the above assumptions and the related analysis for the MTAHV, the resulting control equations of motion in the state space form can be expressed.

$$\dot{x} = Ax + Bu + B_1 \delta_1 \quad (18)$$

The side-slip angle and yaw rate which reflect the stability and handling performance of the vehicle were selected as the control variables. The state vector is

$$x = [\beta_1, \gamma_1, \beta_2, \gamma_2, \beta_3, \gamma_3]^T \quad (19)$$

Control input is

$$u = [\delta_2, \delta_3, \delta_4]^T \quad (20)$$

For simplification, the following forms are first obtained:

$$Mx' = Nx + Ou + P\delta_1 \quad (21)$$

$$A = M^{-1}N$$

$$B = M^{-1}O$$

$$B_1 = M^{-1}P$$

Finally, the matrices M, N, O, P are simplified as follows:

$$M_{11} = m_1 U (S_2 - d_1), M_{12} = I_1, M_{25} = m_3 U S_5,$$

$$M_{26} = -I_3, M_{31} = m_1 U, M_{33} = m_2 U, M_{35} = m_3 U,$$

$$M_{41} = m_1 U S_3, M_{44} = I_2, M_{45} = -m_3 U (S_4 - d_2),$$

$$M_{51} = -U, M_{52} = (S_2 - d_1), M_{53} = U, M_{54} = S_3,$$

$$M_{63} = -U, M_{64} = (S_4 - d_2), M_{65} = U, M_{66} = S_5,$$

$$N_{11} = C_1 (S_1 + S_2 - d_1) - C_2 d_1$$

$$N_{12} = \frac{C_1 S_1 (S_1 + S_2 - d_1) + C_2 S_2 d_1 - m_1 (S_2 - d_1) U^2}{U}$$

$$N_{25} = C_4 (S_5 + S_6)$$

$$\begin{aligned}
N_{26} &= \frac{-C_4(S_5 + S_6)S_6 - m_3S_5U^2}{U} \\
N_{31} &= C_1 + C_2, N_{32} = \frac{C_1S_1 - C_2S_2 - m_1U^2}{U}, N_{33} = C_3, \\
N_{34} &= \frac{-C_3S_4 - m_2U^2}{U}, N_{35} = C_4 \\
N_{36} &= \frac{-C_4S_6 - m_3U^2}{U}, N_{41} = S_3(C_1 + C_2) \\
N_{42} &= S_3 \frac{C_1S_1 - C_2S_2 - m_1U^2}{U}, N_{43} = -C_3S_4 \\
N_{44} &= \frac{C_3S_4S_4}{U}, N_{45} = -C_4(S_4 - d_2) \\
N_{46} &= \frac{(S_4 - d_2)(m_3U^2 + S_6C_4)}{U}, N_{52} = U, \\
N_{54} &= -U, N_{64} = U, N_{66} = -U \\
O &= \begin{bmatrix} d_1C_2 & 0 & 0 \\ 0 & 0 & -(S_6 + S_5)C_4 \\ -C_2 & -C_3 & -C_4 \\ -S_3C_2 & C_3S_4 & (S_4 - d_2)C_4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
P &= \begin{bmatrix} -C_1(S_1 + S_2 - d_1) \\ 0 \\ -C_1 \\ -C_1S_3 \\ 0 \\ 0 \end{bmatrix}
\end{aligned}$$

The unexpressed parts are 0.

2.2. MTAHV Model Constructed in TruckSim

TruckSim is a multi body modelling software with high-precision, which has been widely applied in commercial vehicle virtual simulations (Kim *et al.*, 2016; Keldani and He, 2019). To evaluate the control system, the nonlinear planar model for MTAHV is constructed in TruckSim. Under given or customer-modified road conditions, TruckSim can accurately simulate the vehicle performance by the the

designed controller. TruckSim is applied in this paper to run the MTAHV model with the active multi-axle-steering tractor and full-trailer method proposed to supply the vehicle's dynamic response.

2.3. MTAHV Model Constructed in TruckSim

Aiming to simplify the calculation and control design, enhance the calculation accuracy and minimize the errors caused by the simplification of the multi-axis model, a four-axis B-train double was chosen as a research object. Therefore, the 2A Cab-Over + 1A Flatbed Trl + 1A Flatbed Trl model was selected as the MTAHV in the TruckSim software. 2A Cab-Over + 1A Flatbed Trl + 1A Flatbed Trl model was shown in Figure 2. As shown in Figure 2, the multi-body vehicle model generated using TruckSim software is a B-train double, which is composed of a tractor and two semi-trailers, with neighbouring vehicle units coupled by a fifth-wheel. The non-linear dynamics of relevant components such as fifth wheels, pneumatic tyres, suspension etc. are taken into account. The B-train double model configuration is prescribed as “S_S+S+S”, where “S” indicates a solid axle, the underscore “_” denotes an independent axle groups, and “+” is the fifth-wheel connecting two vehicle units. The steering gear ratio from the steering wheel to the front wheel corner is 25.

The essential parameters of the MTAHV are given in Table 2.



Figure 2. 2A Cab-Over + 1A Flatbed Trl + 1A Flatbed Trl model.

Table 2. Values for the mainly parameters of the MTAHV.

Parameter	Value	Parameter	Value
m_1 (kg)	5760	d_1 (m)	0.9
m_2 (kg)	16640	d_2 (m)	0.1
m_3 (kg)	16640	I_1 (kg·m ²)	41497
S_1 (m)	1.3562	I_2 (kg·m ²)	25933
S_2 (m)	2.5438	I_3 (kg·m ²)	25933
S_3 (m)	4.6280	C_1 (N/rad)	440060
S_4 (m)	3.8720	C_2 (N/rad)	655320
S_5 (m)	4.6280	C_3 (N/rad)	1221500
S_6 (m)	3.8270	C_4 (N/rad)	740400

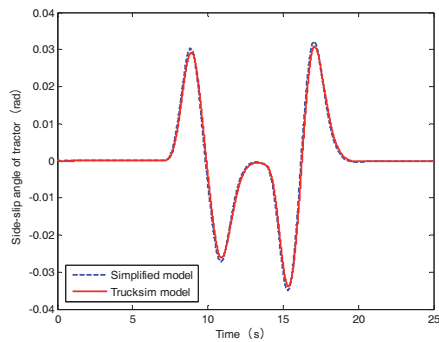
To verify the validity of the simplified model, a standard double lane-change test is performed with the vehicle data presented in Table 2. Simulation results from the linear yaw-plane model with 4-DOF and the nonlinear TruckSim model for the yaw rate, and side-slip angle responses are shown in Figure 3. Figure 3 (a) shows the comparison curve of the tractor's side-slip angle of CG, Figure 3 (b) shows the comparison curve of the first trailer's side-slip angle of CG, Figure 3 (c) shows the comparison curve of the second trailer's side-slip angle of CG. Figure 3 (d) shows comparison curve of tractor yaw rate, Figure 3 (e) shows comparison curve of first trailer yaw rate, Figure 3 (f) shows comparison curve of second trailer yaw rate. As shown in Figure 3, peak values and the tendency of each curve are in good agreement

between the results, so the linear yaw-plane model established meets the following analysis requirements, which provides acceptable dynamic behaviour for the control with high accuracy.

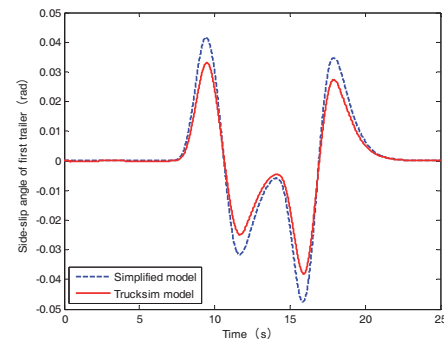
3. ACTIVE MULTI-AXLE-STEERING TRACTOR AND FULL-TRAILER CONTROLLER

3.1. Reference Model

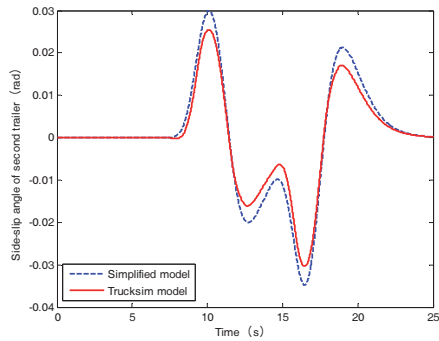
In this section, a reference model supplying the desired state responses is introduced. In the single unit vehicle, to provide familiar scenes for drivers, the actual yaw rate is aimed to track the steady-state yaw rate which is derived from the linear simplified model (Tabatabaei Oreh *et al.*, 2013;



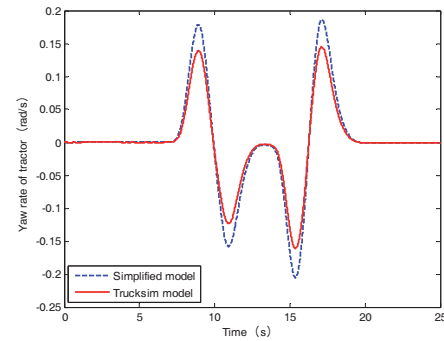
(a) Side-slip angle of tractor



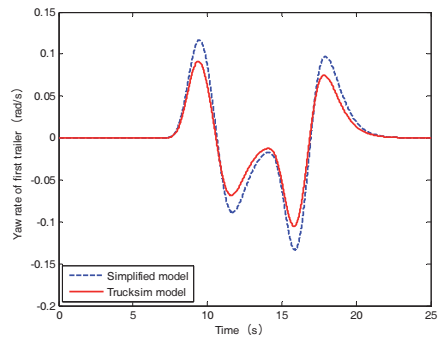
(b) Side-slip angle of the first trailer



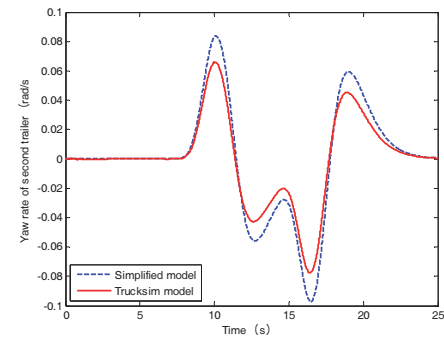
(c) Side-slip angle of the second trailer



(d) Yaw rate of the tractor



(e) Yaw rate of the first trailer



(f) Yaw rate of the second trailer

Figure 3. Comparison curves of state variables of the linear yaw-plane model and TruckSim.

Ghaffari *et al.*, 2011). Similarly, in this paper, to eliminate the off-tracking and make the state variables track the behaviour of desired state, an active control algorithm of multi-axle-steering for the rear axles of the tractor and full-trailer is investigated. In the linear vehicle model, the steady-state yaw rate response is defined as the reference signal for the actual yaw rate of the MTAHV. The lateral velocity indicates the vehicle's stability performance, a zero value has been chosen as the desired value in the single unit vehicle (Boada *et al.*, 2005). Under the assumption of constant forward speed of a vehicle, when the lateral velocity is zero, which is approximately equivalent to zero-side-slip angle. Thus that, side-slip angle as zero may be defined as the desired response for the MTAHV in the steady-state (Wu and Hai, 2003). In this paper, a steady zero side-slip angle is considered as another reference signal for the active steering controller. Therefore, the reference signals for active steering control of MTAHV can be expressed:

$$x_r = [\beta_{1r}, \gamma_{1r}, \beta_{2r}, \gamma_{2r}, \beta_{3r}, \gamma_{3r}]^T \quad (22)$$

Where $\beta_{1r}, \beta_{2r}, \beta_{3r}$ is the reference side-slip angle which sets to zero value, and $\gamma_{1r}, \gamma_{2r}, \gamma_{3r}$ is the steady-state yaw rate of the tractor and two trailer units.

Since in the steady-state condition, the reference states are steady responses of the state vector (Hac *et al.*, 2009).

$$\dot{x}_r = [\dot{\beta}_{1r}, \dot{\gamma}_{1r}, \dot{\beta}_{2r}, \dot{\gamma}_{2r}, \dot{\beta}_{3r}, \dot{\gamma}_{3r}]^T = 0 \quad (23)$$

3.2. Active LQR Multi-axle-steering Controller

In this study, an active LQR multi-axle-steering tractor and full-trailer method is proposed, based on the strategy of tracking the steady-state yaw rate and zero side-slip angle, the block diagram is shown in Figure 3. The active LQR multi-axle-steering tractor and full-trailer method aims to make the tractor and full-trailer keep tracking the desired yaw rate and zero side-slip angle by actively steering the tractor's rear axle, the first trailer's rear axle and the second

trailer's rear axle. The reference model provides the desired states on the basis of the steering angle input of the tractor front axle which is governed by the driver.

The steering angles of the front axle and rear axle of tractor, rear axles of the first trailer and the second trailer are used to make the state variables keep tracking the desired responses presented in Section 3. The block diagram for the active LQR multi-axle-steering tractor and full-trailer method is depicted in Figure 4.

To design the method of active multi-axle-steering tractor and full-trailer based on the strategy of tracking the steady-state yaw rate and zero side-slip angle, the error vector in the tracking process between the actual and reference states is determined as follows:

$$e = \begin{Bmatrix} \beta_1 \\ \gamma_1 \\ \beta_2 \\ \gamma_2 \\ \beta_3 \\ \gamma_3 \end{Bmatrix} - \begin{Bmatrix} \beta_{1r} \\ \gamma_{1r} \\ \beta_{2r} \\ \gamma_{2r} \\ \beta_{3r} \\ \gamma_{3r} \end{Bmatrix} = x - x_r \quad (24)$$

The reference state vector can be formulated as

$$\dot{x}_r = Ax_r + B_1\delta_1 \quad (25)$$

Since Equation (23), the reference states can be obtained as follows:

$$x_r = [\beta_{1r}, \gamma_{1r}, \beta_{2r}, \gamma_{2r}, \beta_{3r}, \gamma_{3r}]^T = -A^{-1}B_1\delta_1 \quad (26)$$

To design the LQR controller, the following equation can be achieved by replacing the term at the right side of Equations (18) and (25) with the derivative value of the tracking error:

$$\dot{e} = \dot{x} - \dot{x}_r = A(x - x_r) + Bu = Ae + Bu \quad (27)$$

The active LQR multi-axle-steering tractor and full-trailer method based on the strategy of tracking the desired state response, which aims to make the tractor and full-trailer keep tracking the steady-state yaw rate and zero side-slip angle of the CG at the same time. So that, the cost function can be expressed:

$$J = \int_0^\infty \left[q_1\beta_1^2 + q_2(\gamma_1 - \gamma_{1d})^2 + q_3\beta_2^2 + q_4(\gamma_2 - \gamma_{2d})^2 + q_5\beta_3^2 + q_6(\gamma_3 - \gamma_{3d})^2 + r_1\delta_1^2 + r_2\delta_2^2 + r_3\delta_3^2 \right] dt \quad (28)$$

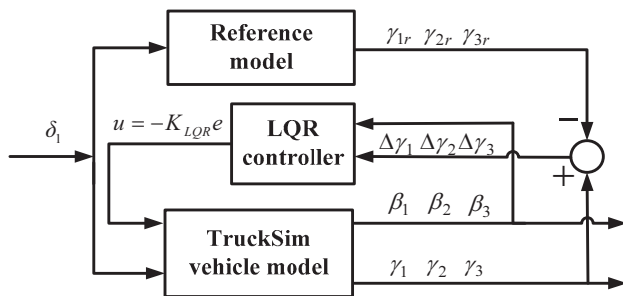


Figure 4. Block diagram for the active LQR multi-axle-steering tractor and full-trailer method.

With the control input and state vectors, the above performance indexes provided by Equation (28) can also be expressed in a standard form:

$$J = \int_0^{\infty} (e^T Q e + u^T R u) dt \quad (29)$$

Where $Q = \text{diag}(q_1, q_2, q_3, q_4, q_5, q_6)$, and $R = \text{diag}(r_1, r_2, r_3)$.

Since the states of the 4-DOF yaw-plane model can be directly measurable, they can be used as the foundation of the feedback. The control signal can be expressed as the linear combination of state variables, in the issue of direct state feedback:

$$u = -K_{LQR} e \quad (30)$$

Where u represents one of the controlled variables. Where K_{LQR} is called the feedback matrix, which is figured out by the positive (semi-)definite solution of the Algebraic Riccati equation:

$$E_{LQR} A + A^T E_{LQR} - E_{LQR} B R^{-1} B^T E_{LQR} + Q = 0 \quad (31)$$

And K_{LQR} can be constituted as follows:

$$K_{LQR} = R^{-1} B^T E_{LQR} \quad (32)$$

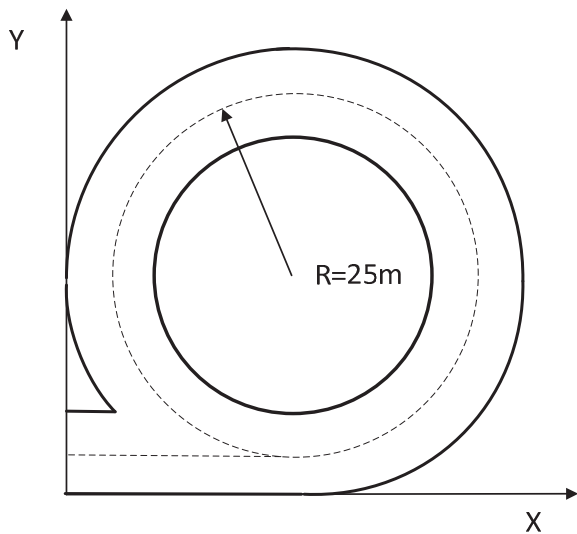


Figure 5. 360° roundabout test.

4. SIMULATION RESULTS

To assess the lateral stability of the MTAHVs with the proposed active multi-axle-steering method, two test manoeuvre are carried out through Matlab/Simulink and TruckSim co-simulation. The first is the 360° roundabout path-following test manoeuvre at low speed (Tsao *et al.*, 2006; Edgar, 2004), and the second is the single-lane change test manoeuvre at high speed (ISO 14971, 2000), which are generally adopted for evaluating low-speed PFOT and high-speed RWA ratio of the MTAHVs, respectively. The two test manoeuvres are shown in Figures 5 and 6, respectively.

4.1. 360° Roundabout Manoeuvre

At the low-speed condition of 360° roundabout manoeuvre, the center of the front axle of tractor is required to drive past a straight section first and then to follow a roundabout road with a radius of 25 m. The low-speed maneuverability of the MTAHV is examined based on the circular test with the 10 km/h forward speed. The PFOT is used to assess the active steering strategy performance at low speeds: The final weighting coefficient for low-speed circular condition is selected: $q_1 = 0.5$, $q_2 = 0.05$, $q_3 = 2$, $q_4 = 0.05$, $q_5 = 2$, $q_6 = 0.05$; $r_1 = 1$, $r_2 = 1$, $r_3 = 1$.

Figure 7 illustrates the trajectories of the centers of the multi-axle in the 360° circle test before and after using the active LQR multi-axle-steering control, respectively. As shown in Figure 7, with the active multi-axle-steering method, the PFOT between the second trailer and the tractor changes from 3.35 m to 2.13 m, which is reduced by 36.4 %; while the PFOT between the first trailer and the tractor changes from 1.76 m to 1.04 m, which is decreases by 40 %. Table 3 lists the performance measures PFOT under the low-speed 360° manoeuvre. It can be indicated from the simulation results that the axle centers of the first and second trailers follow closely the tractor's front axle center. Furthermore, the maximum radial offset between the path of the multi-axle centers and that of the desired path has also been decreased to varying degrees, the occurrence of inward movement of the MTAHV has been limited. The maneuverability improvement is directly due to the application of the active LQR multi-axle-steering method.

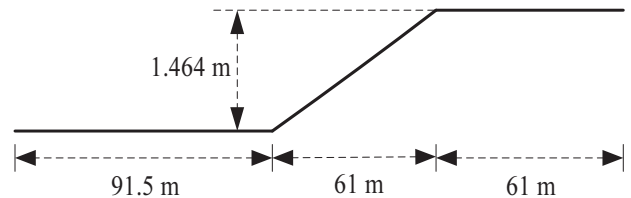


Figure 6. Single lane change test.

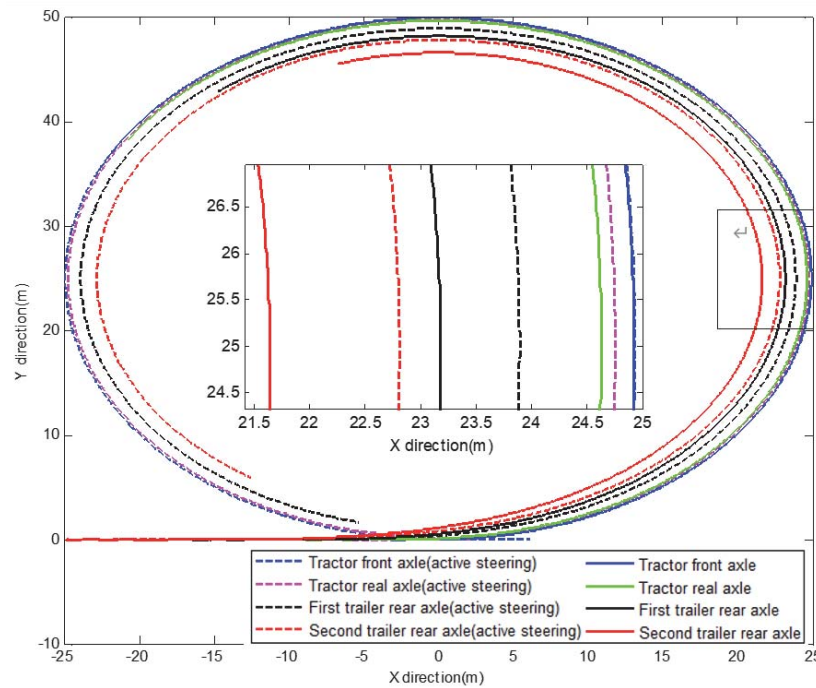


Figure 7. Trajectories of the centers of the multi-axle in the 360° roundabout test before and after the control.

Table 3. Performance measures (PFOT) under the low-speed 360° manoeuvre.

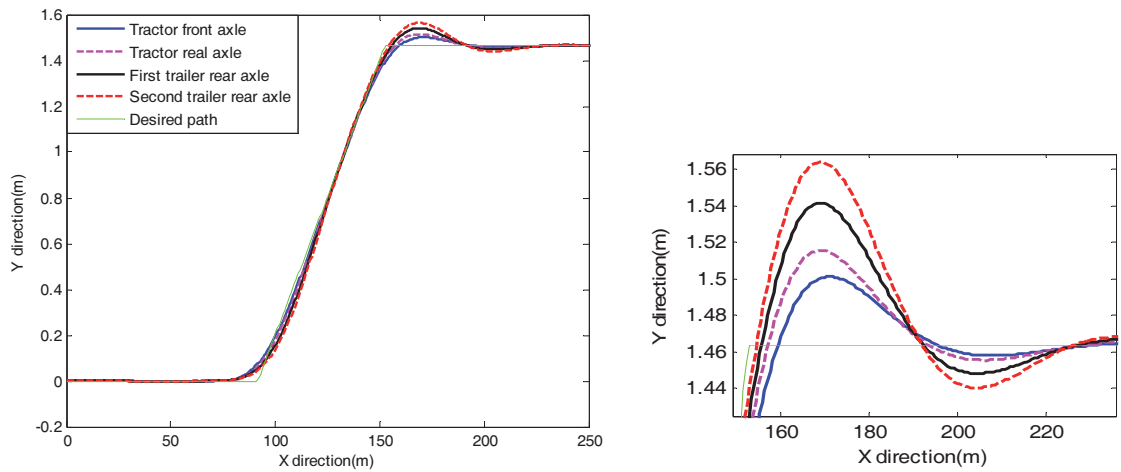
Performance Measures		Without control	With the active steering control	Reduction ratio
PFOT/m	First trailer	3.35	1.76	40%
	Second trailer	2.13	1.04	36.4%

4.2. High-Speed Single-Lane-Change Test

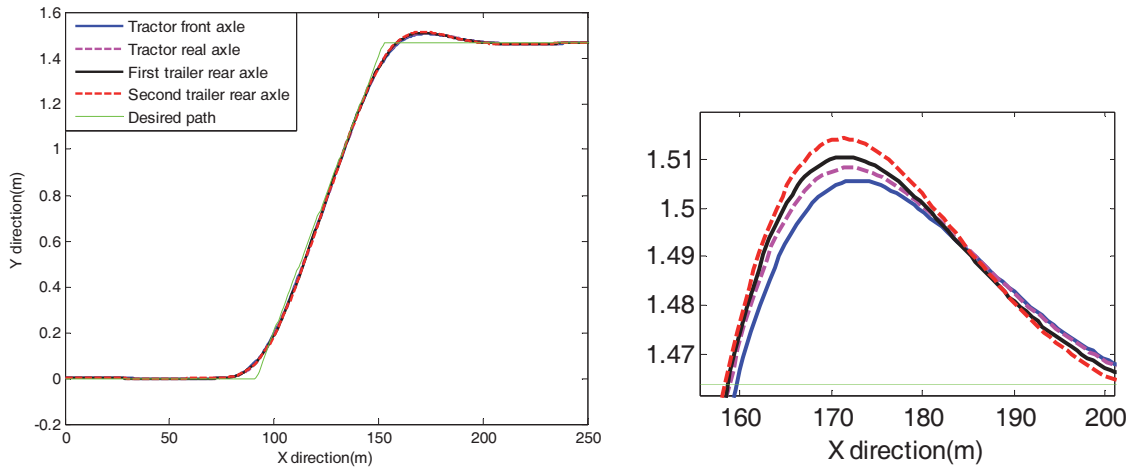
To investigate the high-speed stability of the MTAHV, the single-lane change test is simulated. The methodologies specified in ISO 14791 require that the articulated heavy vehicles perform a single-lane change manoeuvre at a speed of 88 km/h (ISO 14791, 2000), which are commonly adopted for evaluating high-speed lateral stability of the MTAHVs (Preston-Thomas and El-Gindy, 1998; Zhu *et al.*, 2021; Wang and He, 2016; Islam *et al.*, 2012). The MTAHV is driving at a forward speed of 88 km/h on a straight-path segment. Then, a sudden obstacle avoidance lane-change is carried out. The final weighting coefficient for high-speed lane change condition is selected: $q_1 = 42$, $q_2 = 0.1$, $q_3 = 42$, $q_4 = 0.1$, $q_5 = 48$, $q_6 = 0.1$; $r_1 = 1$, $r_2 = 1$, $r_3 = 1$. Under high-speed single-lane-change test, the active LQR multi-axle-steering method is also useful to enhance the handling stability under high-speed conditions. It has been witnessed that articulated heavy vehicles may display dangerous motion while turning at a high speed (He *et al.*, 2010). The high-speed directional performance is more associated with its complicated structure according to its large size. Figures

10 ~ 20 show the simulation results under high-speed single-lane-change manoeuvre.

Figure 8 provide very useful simulation conclusions for researching whether the units of the MTAHV can follow the center trajectory of the tractor's front axle accurately in a high-speed obstacle avoidance test. Without active steering control, there are a large off-tracking between the rear axle trajectory centers of the three units and the tractor's front axle trajectory center at the beginning of the turn and return phase. Simulation indicates that the method proposed provides a suitable level of tracking performance to both the trajectory of the tractor's front axle and the desired trajectory. With the application of the active LQR full-axle-steering method, the center of the tractor's rear axle follows the center path of the tractor's front axle more accurately, with the maximum transient off-tracking of 0.002 m, decreasing 86.7 % from the corresponding original value of 0.315 m. The first trailer's rear axle center tracks the center trajectory of the tractor's front axle accurately with the peak transient off-tracking of 0.04 m, reducing 2.4 % from the value of 0.41 m. The second trailer's rear axle center tracks the center trajectory of the



(a) Without active LQR steering control



(b) With active LQR steering control

Figure 8. Trajectories of the center of the multi-axle of the MTAHV in a single lane change test.

tractor's front axle more accurately, the transient off-tracking has been reduced from 0.063 m to 0.008 m, which is a 87.3 % reduction. Results shown in Figure 8 indicate that using the method designed, the MTAHV has higher handling stability and the trailer can more accurately track the tractor's path under high-speed obstacle avoidance condition.

In Figures 9, 10, 11, the time responses of the side-slip angle of three unit's CG are depicted, respectively. The baseline vehicle has relatively poor handling stability, since the curves of side-slip angle of CG of three units fluctuate and exhibit very greatly variations with greater peak.

In Figures 9, 10, 11, when the MTAHV is fitted with the active LQR multi-axle-steering method proposed, the side-slip angle responses of the three units have been strongly enhanced, respectively, which keep to a relatively smooth curve, and tend to converge rapidly. The maximum side-slip angle of the tractor CG has been dropped by 74.4 %, from

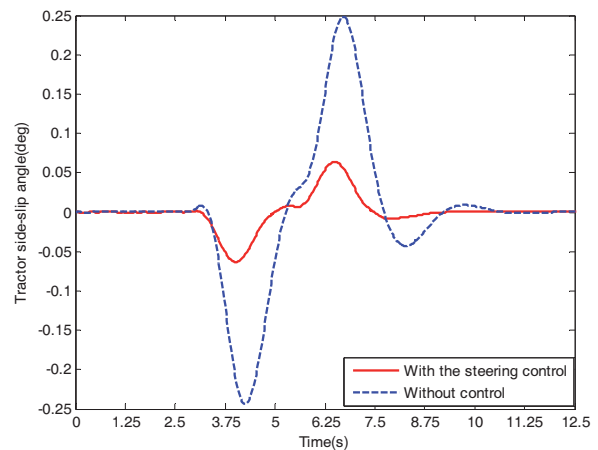


Figure 9. Time history of side-slip angle of the tractor's body.

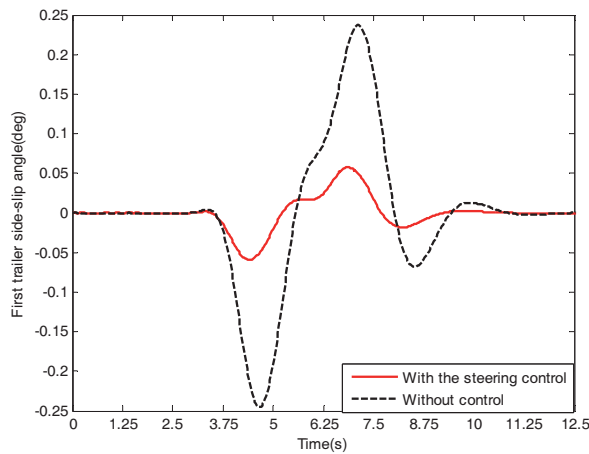


Figure 10. Time history of side-slip angle of the first.

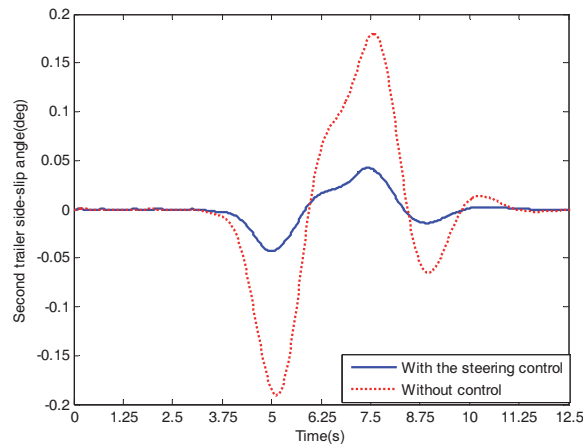


Figure 11. Time history of side-slip angle of the second trailer's body.

0.2485 deg to 0.06354 deg, while that of the first trailer CG has been decreased from 0.245 deg to 0.05744 deg, with a reduction of 76.6 %, and the active steering strategy decreases the peak value of the side-slip angle of the CG of the second trailer by 77.7 %, from 0.1907 deg to 0.04259 deg.

In the high-speed obstacle avoidance lane change manoeuvre, the articulation angles between the connected units of the MTAHV are also studied. In Figures 12, 13, the time responses of the articulated angles at three unit's CG are depicted, respectively. For the active LQR full-axle-steering controlled MTAHV, the articulation angle between the tractor and the first trailer is 0.4725° , decreasing by 25.4 % from the original value of 0.6331° . Simultaneously, the articulation angle between the first trailer and the second trailer is 0.3562° , which is reduced by 23.7 % from the initial value of 0.4663° . The reduction of articulation angles indicates that the probability of jackknifing phenomenon of the MTAHV will be greatly reduced.

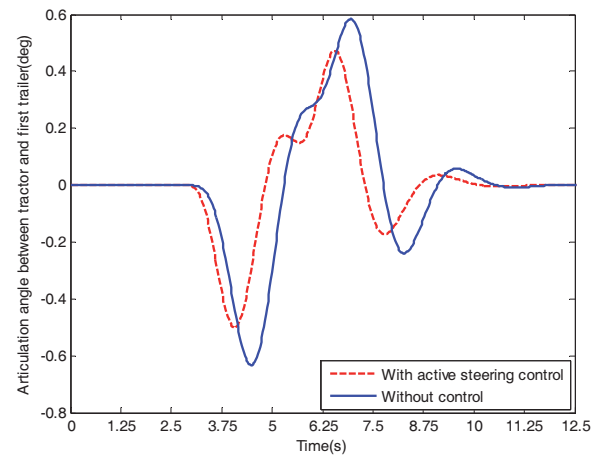


Figure 12. Time history of articulation angle between the first trailer and the tractor.

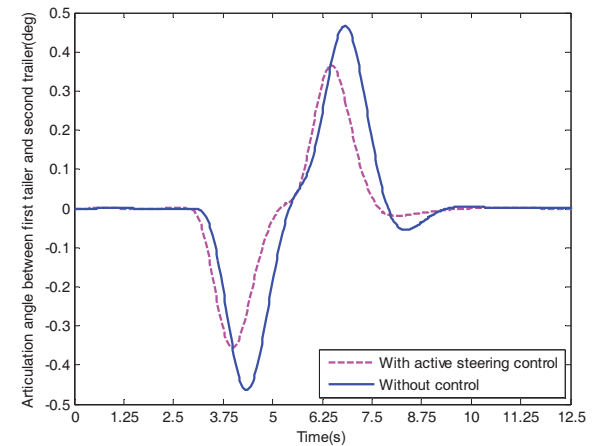


Figure 13. Time history of articulation angle between two trailers' units.

In Figures 14, 15, and 16, the yaw rate responses of the three units can be observed. With the application of the active steering strategy, peak yaw rate of the tractor, the first trailer, and the second trailer have been diminished from 1.622 deg/s to 1.316 deg/s with a reduction of 18.9 %, from 1.572 deg/s to 1.3 deg/s with a 17.3 % reduction, and from 1.578 deg/s to 1.287 deg/s by as much as 18.4 %, respectively. It is noted that the active steering strategy proposed strongly improves the yaw rate responses of the three units. The reduction in the yaw rate implies that the probability of lateral sliding of its rear axle is limited at high speeds.

The total lateral tire forces of the rear axles of the three units are illustrated in Figures 17, 18 and 19. The total lateral tire force on the tractor's rear axle was reduced from 4,698 to 4,348 N, which is a 7.4 % reduction, that of the first trailer was diminished by 8.6 % from 9,998 N to 9,143 N, and that of the second trailer was decreased from 5,663 to 4,875 N,

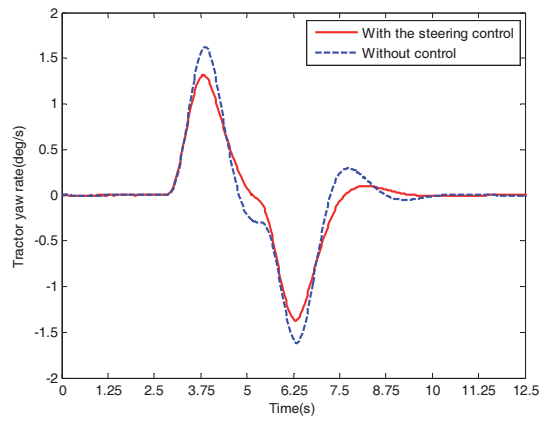


Figure 14. Time history of tractor yaw rate.

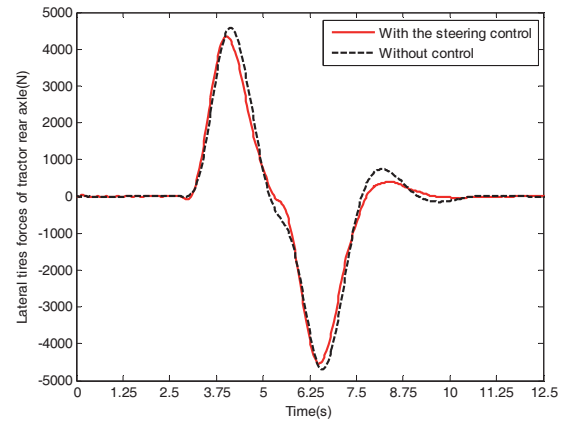


Figure 17. Time history of total lateral tire force of tractor rear axle.

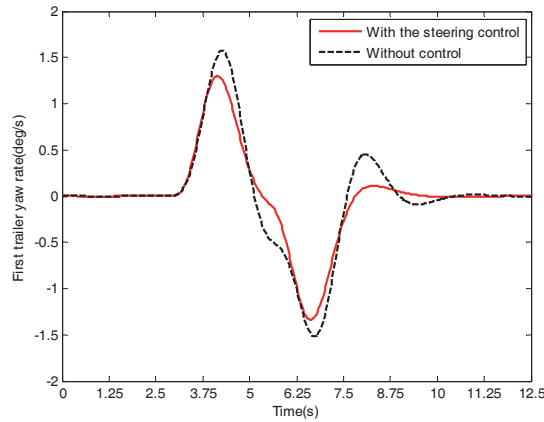


Figure 15. Time history of first trailer yaw rate.

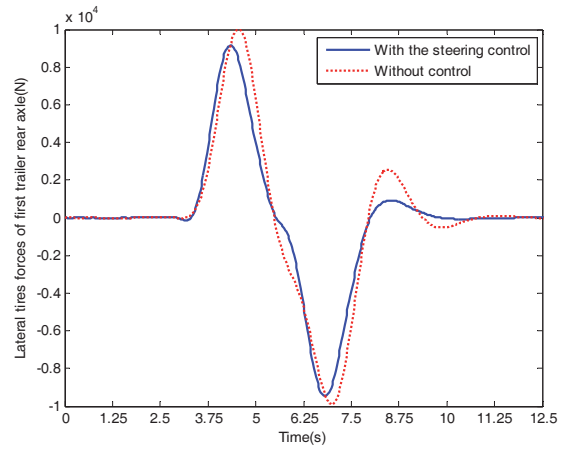


Figure 18. Time history of total lateral tire force of first trailer rear axle.

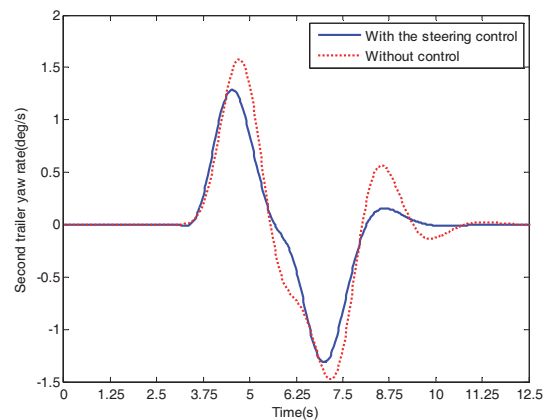


Figure 16. Time history of second trailer yaw rate.

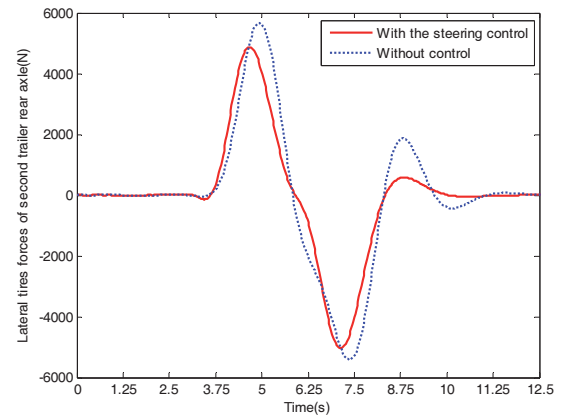


Figure 19. Time history of total lateral tire force of second trailer rear axle.

which is a reduction of 14 %. The reduction in the total lateral tire force of rear axles of the MTAHV improved effectively the tire wear characteristics in the obstacle avoidance single-lane change manoeuvre.

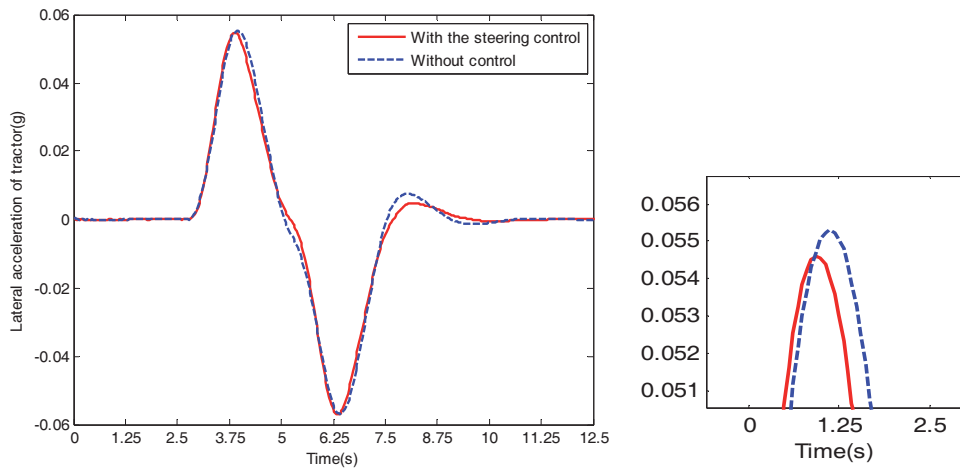


Figure 20. Time history of tractor's lateral acceleration.

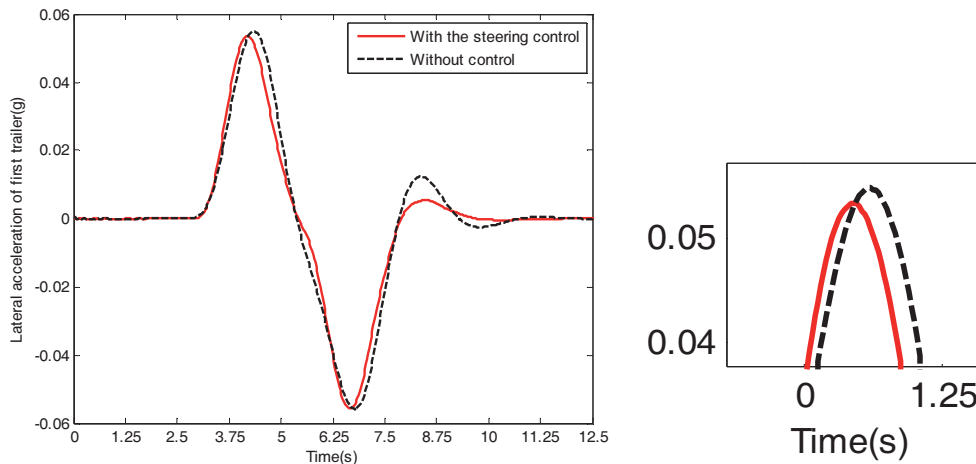


Figure 21. Time history of the first trailer's lateral acceleration.

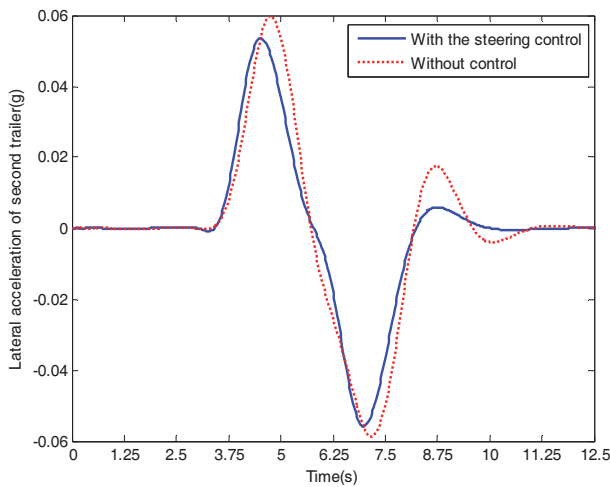


Figure 22. Time history of the second trailer's lateral acceleration.

Figures 20, 21, and 22 show the improvement of the three units' lateral acceleration with the method designed. The active LQR multi-axle-steering method has a weaker effect on the lateral acceleration of the tractor and the first trailer, but the second trailer is most affected by the method proposed. The rearmost trailer has higher lateral acceleration than that of the tractor, which contributes to the high risk of trailer rollover. Therefore, it is necessary to control the rearmost trailer's lateral acceleration. The tractor's lateral acceleration is dropped by 1.2 %, that of the first trailer is reduced up to 4.2 %, and that of the second trailer is diminished by 10.4 %, from 0.05988 to 0.05364 g. The lateral acceleration of the three units with the strategy designed has been reduced and smoothed, which indicates that the anti-roll control is enhanced.

The RWA ratio is recognized as an important performance measurement index, which is mainly used to evaluate the high-speed lateral stability of the MTAHVs. According to

the definition of the RWA ratio and Figures 20 ~ 22, with the application of the strategy proposed, the RWA ratio of the first trailer is decreased by 3.1 %, from 1.01 to 0.979, and that of the second trailer is diminished from 1.083 to 0.9822, which is 9.3 % reduction. The size of the RWA ratio is key for the stability of a heavy vehicle combination. A lower RWA ratio indicates that the semi-trailers undergo lower levels of lateral acceleration during frequency-dependent oscillations or transient manipulation (Ding *et al.*, 2013). Consequently, a lower value of lateral acceleration of the semi-trailer will yield the better lateral stability for the MTAHVs. The reduction in the RWA ratio distinctly implies that the improvement in the directional behavior of the MTAHV can be realized by using the active LQR steering method proposed.

To summarize the simulated high-speed single lane change test for the MTAHV using the active LQR multi-axle-steering method, Table 4 lists the quantitative simulation results. The reduction of the PFOT enhances the ability to keep following roundabouts closely at low speed. In the case of high speed condition, the diminishing of the transient off-tracking yields more accurately path-following, the reduction of the peak side-slip angles of three units yields higher handling stability, the reduction of articulation angles

indicate that the probability of jackknifing phenomenon of the MTAHV is reduced, the reduction in the yaw rate implies that the probability of lateral sliding of its rear axle is limited, the reduction in the total lateral tire force of rear axles improves effectively tire wear characteristics in the obstacle avoidance manoeuvre, the lateral acceleration of the three units have been reduced and smoothed, which indicates that the anti-roll control is enhanced, and the reduction of the RWA ratio distinctly implies that the improvement in the directional behavior can be realized.

5. CONCLUSION

This paper focused on the active LQR multi-axle-steering method to improve the maneuverability and stability of the MTAHV. The following conclusions can be drawn:

- (1) The linear yaw-plane model with 4-DOF for the MTAHV has been constructed and validated, the linear yaw-plane model provides a dependable basis for designing the reference model and the control system.
- (2) The active control algorithm of multi-axle-steering for the rear axles of the tractor and full-trailer is investigated, and a LQR controller is proposed on the basis of the linear yaw-plane model with 4-DOF for the MTAHV to

Table 4. Performance measures in the single lane change test at high speed.

Performance measures		Without control	WITH the active steering control	Reduction ratio
Transient off-tracking (m)	First trailer rear axle	0.015	0.002	86.7 %
	First trailer rear axle	0.041	0.04	2.4 %
	Second trailer rear axle	0.063	0.008	87.3 %
Absolute peak value of side-slip angle (deg)	tractor	0.2485	0.06354	74.4 %
	First trailer	0.245	0.05744	76.6 %
	Second trailer	0.1907	0.04259	77.7 %
	tractor	1.622	1.316	18.9 %
Yaw rate (deg/s)	First trailer	1.572	1.3	17.3 %
	Second trailer	1.578	1.287	18.4 %
Absolute peak value of articulated angle (deg)	between tractor and first trailer	0.6331	0.4725	25.4 %
	between two trailers	0.4663	0.3562	23.7 %
Absolute peak value of lateral tires forces (N)	Tractor rear axle	4698	4348	7.4 %
	First trailer rear axle	9998	9143	8.6 %
	Second trailer rear axle	5663	4875	14 %
Absolute peak value of lateral acceleration (g)	Tractor	0.0553	0.05461	1.2 %
	First trailer	0.05583	0.0535	4.2 %
	Second trailer	0.05988	0.05364	10.4 %
RWA ratio	First trailer	1.083	0.9822	9.3 %
	Second trailer	1.01	0.979	3.1 %

make the control variables follow the desired state responses.

- (3) The important effectiveness of the approach proposed on enhancing the maneuverability and stability of a MTAHV have been validated through the simulations of the low-speed 360° roundabout trajectory tracking test and the high-speed lane-change maneuver, respectively.
- (4) It is found that the active LQR multi-axle-steering method has more advantages in improving the high-speed handling respond and lateral performance, effectively reducing the PFOT, the transient off-tracking, peak values of side-slip angle of CG, yaw rates, articulation angles of the adjacent units, the total lateral tire force of rear axles, lateral acceleration, and RWA ratio. The method has a certain reference value for improving the active safety of the MTAHV.

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